

FastWeigh / TAC Insight — Call Bullets

Audience: Brad Artis (TAC Insight / FastWeigh sales) · **For:** David Smith, Dividia · **Date:** 2026-05-27

Goal: close out the technical unknowns that block the Jahna quote. Anything answered here removes a conditional from the quote.

Context Brad needs up front: FastWeigh is not yet installed at Jahna — TAC Insight is 2-3 weeks out, replacing the current Mettler Toledo DataBridge MS ticketing system. We are synchronizing our delivery to FastWeigh go-live week, so two migrations happen concurrently: (1) Jahna swaps DataBridge MS → FastWeigh (their project), and (2) we replace AxIEye + the DataBridge AxIEye/file-rename plugins with our own NVR + Jetson stack (our project). Our piece handles QR permit reading, axle measurement, bridge-formula validation, and the cloud audit-trail portal. FastWeigh owns ticketing and the loader-tablet app going forward.

Top 3 — these block the quote

BLOCKER 1. Ticket-in-progress event (or query)

We need to attach our measurement + permit to the correct ticket *before* it prints. We have seen `TicketPending` in the GraphQL schema.

Is there a webhook event when a ticket enters the in-progress state — `ticket.pending`, `ticket.started`, `ticket.tare_recorded`? If not, can we query `TicketPending` via GraphQL on demand the moment our QR scan fires?

Why it matters: without this, we cannot reliably associate measurement data with a specific ticket while the truck is still on the scale. Drives the entire correlation design.

If yes (webhook exists): simplest path — we listen + attach. **If only GraphQL:** we poll on QR scan, add ~10–20 hr to Phase 0. **If neither:** we anchor on our own scale-on event (recorder already has `Scale_TruckOn`) and correlate by truck + time window — adds ~20–40 hr.

BLOCKER 2. Tare ↔ loaded ticket linking

What field connects a tare ticket to its later loaded ticket?

- `truck_id`, `slip_number`, `transaction_guid`, or other?
- Does the loaded ticket carry an explicit reference to the tare, or is it correlation-by-truck-within-time-window?
- Are tare + loaded the same event type, or distinct (e.g., `ticket.tare` vs `ticket.loaded`)?
- If same event, what field tells us which step we're on? (`tare_amount` populated vs `gross_amount`? A `ticket_step` enum?)

Why it matters: drives "paired mode" — merging tare measurement + loaded weight into one cloud audit record. Three possible models on the table (A: separate tickets with parent reference; B: one ticket with state transitions; C: no tare ticket, only loadout). Our pipeline works in any, only the correlation code changes.

BLOCKER 3. Webhook reliability semantics

Critical for lightning-grade reliability tier. Need answers to all four:

- **Retry policy** — Svx exponential backoff by default. Confirm window, max attempts.
- **Idempotency key on the payload** — does each delivery carry a stable ID so we can dedupe on our side?
- **Replay-on-outage** — if our cloudapi is down for an hour, will FastWeigh resend missed deliveries? Or do those tickets go un-attached forever?

- **Out-of-band reconciliation** — is there a REST/GraphQL endpoint we can poll for "all tickets since timestamp X" so we can sweep up anything missed?

Why it matters: if webhooks are fire-and-forget with no replay, the reliability tier needs a Dividia-side reconciliation worker (~30–60 hr). If FastWeigh replays missed deliveries and the payload is idempotent, the reliability tier shrinks to monitoring + UPS spec.

Tier 2 — these shape line items in the quote

HIGH 4. Custom UDFs in webhook payloads

Multi-site routing problem. Jahna has 6 sites — each has its own NVR.

- Can we add custom User-Defined Fields to the webhook body (site code, NVR serial)?
- If not, can the Svix endpoint URL carry query parameters per site?
- If neither, what does TAC Insight recommend for multi-site routing? (Per-site signing secret? Per-site endpoint URL?)

Why it matters: the multi-site rollout (6 sites for Jahna) needs a clean routing pattern. Without it we kludge with substring matching on site codes inside ticket fields, which is fragile.

HIGH 5. Loader-tablet app — what does it surface?

Today Jahna runs a custom HTML poller on the loader tablets that reads from the scale-PC folder. That sunsets when FastWeigh's loader app comes online. We need to know what the FastWeigh loader app actually displays so we know whether our measurement / allowed-gross / permit-valid status reaches the operator.

- Does the FastWeigh loader-tablet app have a field for "max allowed weight from external measurement"?
- Can we push that value into FastWeigh via API for the loader to read?
- Or does the operator only see what FastWeigh natively computes, and our number lives in our portal only?

Why it matters: if FastWeigh can't surface our measurement to the loader, we either keep a Dividia-side tablet display (adds ~40–60 hr to Phase 1) or push hard for them to add the field. Material to scope.

HIGH 6. Sample payloads + sandbox

- Redacted sample webhook payload for tare and for loaded, actual JSON (not just field list).
- A sandbox / dev FastWeigh account we can develop against while Jahna's tier upgrade is pending.

Why it matters: we are building blind until we see real shapes. Sandbox unblocks weeks of work and removes guesswork from the quote.

Tier 3 — context-setting and cost

MED 7. Subscription tier

Jahna doesn't have webhook / GraphQL access enabled. What tier do they need to upgrade to, and what's the cost? Include this in our budgetary quote so they see the all-in number.

MED 8. Scale events from FastWeigh (nice-to-have)

Does FastWeigh emit webhooks for scale events (truck entering, truck on scale, truck leaving), or only completed tickets? Real-time scale weight API? *Not blocking* — our recorder has its own scale-sensing path — but if FastWeigh provides it cleanly we'd avoid a parallel scale tap on some sites.

What we are NOT asking (already known)

- FDOT permit lookup — already implemented. Public QR URL, no credentials, verified against real permit. Not a FastWeigh dependency.
- Scale hardware — Jahna's scale indicators are SERIAL-output (Mettler Toledo IND780 assumed, not 100% confirmed). Brought onto IP via Moxa NPort 5110 serial-to-Ethernet (1 per inbound + 1 per outbound on a typical 2-scale site). Quote 17938 includes 2 × NPort 5110 for this. **[CORRECTED 2026-06-09 — David]** the prior "IP-based, no Moxa needed" was wrong.
- Whether AxIEye stays — no, we replace it. Jahna confirmed full replacement.
- Whether DataBridge stays — no, we replace its remaining functions (file rename, InYard, tablet feed) as part of Phase 1.

Decision matrix — what Brad's answers do to our quote

Brad's answer	Quote impact
Ticket-in-progress webhook exists	Phase 0 stays at low end (~15 hr). Simplest correlation model.
Only GraphQL TicketPending query	Phase 0 +10–20 hr. We add a polling worker triggered by QR scan.
Neither — we self-anchor	Phase 0 +20–40 hr. Scale-on event drives transaction; FastWeigh ticket joins later by time window + truck.
Webhooks have full replay on outage + idempotency keys	Reliability "Standard" tier is enough. Monitoring + UPS only.
Webhooks are fire-and-forget	Reliability "Failover" tier adds a reconciliation worker (~30–60 hr) that polls REST/GraphQL for missed tickets.
UDFs supported	Clean multi-site routing. No additional cost per site after #2.
No UDFs, only per-site endpoint	+2–4 hr per site for endpoint provisioning. Tolerable.
Loader app can surface external measurement	Phase 1 base price holds. Our measurement reaches the operator via FastWeigh.
Loader app cannot surface our data v1	Phase 1 +40–60 hr for a Dividia loader display (web or native), OR we hold the loader display unchanged on the existing HTML poller until FastWeigh adds the field.
Sandbox available now	Phase 0 runs in parallel with Jahna's FastWeigh install. No idle time.
No sandbox	Phase 0 work waits until Jahna's go-live. Calendar risk to Van Fleet pilot date.

Cross-references: jahna-fastweigh hub at [cloudapi/docs/jahna-fastweigh/open-questions.md](#) · [transaction-lifecycle-deep-dive.md](#) · [architecture.md](#) § 0c (current operational flow).